
Preface

The following preface is reprinted from the model National Building Code of Canada (NBC), on which the British Columbia Building Code (BCBC) is substantially based. As such, this information also pertains to the BCBC. Material that is not relevant to the BCBC has been deleted; minor BCBC-specific edits within the text have been made. Additional information only relevant to the BCBC has been included in text boxes within the document.

The National Building Code of Canada 2020 (NBC), together with the National Plumbing Code of Canada 2020 (NPC), the National Fire Code of Canada 2020 (NFC) and the National Energy Code of Canada for Buildings 2020 (NECB), has been developed by the Canadian Commission on Building and Fire Codes (CCBFC) as an objective-based national model code that can be adopted by provincial and territorial governments.

In Canada, provincial and territorial governments have the authority to enact legislation that regulates building design and construction within their jurisdictions. This may involve the adoption of the NBC without change or with modifications to suit local needs, and the enactment of other laws and regulations regarding building design and construction, including requirements for professional involvement.

In British Columbia, the NBC, together with the National Plumbing Code of Canada, are adopted, including any variations considered necessary, as Book I (General) and Book II (Plumbing Systems) of the BCBC, pursuant to the Building Act. The National Fire Code of Canada is adopted, including any variations considered necessary, as the British Columbia Fire Code (BCFC), pursuant to the Fire Services Act. The National Energy Code of Canada for Buildings (NECB) is adopted by reference from Book I (General) of the British Columbia Building Code.

The NBC is a model code in the sense that it helps promote consistency among provincial and territorial building codes. Persons involved in the design or construction of a building should consult the provincial or territorial jurisdiction concerned to find out which building code is applicable.

This edition of the NBC succeeds the 2015 edition.

This edition of the BCBC succeeds the 2018 edition.

Development of the National Model Codes

The CCBFC, an independent committee established by the National Research Council of Canada (NRC), is responsible for the content of the National Model Codes. The CCBFC is made up of volunteers from across the country and from all facets of the Codes-user community. Members of the CCBFC and its standing committees include builders, engineers, skilled trade workers, architects, building owners, building operators, fire and building officials, manufacturers, and representatives of general interests.

The CCBFC is advised on scope, policy and technical issues pertaining to the Codes by the Provincial/Territorial Policy Advisory Committee on Codes (PTPACC), which is a committee of senior representatives from provincial/territorial ministries responsible for building, fire, plumbing and energy regulation in their jurisdictions. The PTPACC was created by the provinces and territories, with provision of guidance to the CCBFC as one of its main functions. Through the PTPACC, the provinces and territories are engaged in every phase of the Codes development process.

Codes Canada staff within the Construction Research Centre at the NRC provide technical and administrative support to the CCBFC and its standing committees, and coordinate the provision of evidence-based research to inform Codes development. The NRC publishes the National Model Codes and periodic revisions to the Codes to address pressing issues.

Similar to the Codes themselves, such periodic revisions do not have legal effect until adopted into law by the Province of British Columbia.

The broader Codes-user community also makes significant contributions to the Codes development process by submitting requests for changes or additions to the Codes and by commenting on the proposed changes during the public reviews that precede each new edition.

The CCBFC takes into consideration the advice received from the provinces and territories as well as Codes users' comments at each stage of Codes development. The scope and content of the National Model Codes are determined on a consensus basis, which involves the review of technical, policy and practical concerns and discussion of the implications of these concerns.

More information on the Codes development process is available on the NRC's website.

National Building Code of Canada 2020

The NBC sets out technical provisions for the design and construction of new buildings. It also applies to the alteration, change of use and demolition of existing buildings.

The NBC establishes requirements to address the following five objectives:

- safety
- health
- accessibility
- fire and structural protection of buildings
- environment

Code provisions do not necessarily address all the characteristics of buildings that might be considered to have a bearing on the Code's objectives. Through the extensive consensus process used to develop and maintain the National Model Codes (see the section entitled Development of the National Model Codes), the Codes-user community has decided which characteristics should be regulated through the NBC.

The provisions of the NBC can be considered as the minimum acceptable measures required to adequately achieve the above-listed objectives, as recommended by the CCBFC. Once they are adopted into law or regulation by an authority having jurisdiction, the provisions become minimum acceptable requirements representing the minimum level of performance required to achieve the objectives that is acceptable to the adopting authority.

In context of the preceding paragraph, the terms "authority having jurisdiction" and "adopting authority" refer to the government of the Province of British Columbia. This differs from the context of application and enforcement of the BCBC, for which the Code contains a distinct definition of the term "authority having jurisdiction". (See Article 1.4.1.2. of Division A.)

The NBC is a model code which, when adopted or adapted by a province or territory, becomes a regulation. It is not a guideline on building design or construction. The design of a technically sound building depends upon many factors beyond compliance with building regulations. Such factors include the availability of knowledgeable practitioners who have received appropriate education, training and experience and who are familiar with the principles of good building practice and experience using reference manuals and technical guides.

The NBC does not list acceptable proprietary building products. It establishes the criteria that building materials, products and assemblies must meet. Some of these criteria are explicitly stated in the NBC while others are incorporated by reference to material or product standards published by standards development organizations. Only those portions of the standards related to the objectives of this Code are mandatory parts of the NBC.

Relationship between the NBC and the NFC

The NBC and NFC each contain provisions that relate to the safety of persons in buildings in the event of a fire and the protection of buildings from the effects of fire.

These two model codes are developed as complementary and coordinated documents to minimize the possibility of their containing conflicting provisions. It is expected that buildings comply with both the NBC and the NFC.

The NBC covers the fire safety and fire protection features that are required to be incorporated in a building at the time of its original construction. Building codes typically no longer apply once a building is occupied, unless the building is undergoing alteration or change of use, or being demolished.

The NFC covers:

- the ongoing maintenance and use of the fire safety and fire protection features incorporated in buildings
- the conduct of activities that might cause fire hazards in and around buildings
- limitations on hazardous contents in and around buildings
- the establishment of fire safety plans
- fire safety at construction and demolition sites

In addition, the NFC contains provisions regarding fire safety and fire protection features that must be added to existing buildings when certain hazardous activities or processes are introduced into these buildings.

Some of the NFC's provisions are incorporated by reference in the NBC and, thus, may apply to original construction, alterations, or changes in use.

Relationship between the NBC and the NECB

The provisions in Section 9.36. of Division B of the NBC are tied to the environment objective. These provisions, which apply to housing and small buildings, have a similar scope to that of the NECB, except that they do not address lighting and electrical power systems. The NECB is referenced in NBC Section 9.36. as an acceptable solution.

Code Requirements

Most NBC requirements address at least one of the Code's five stated objectives (safety, health, accessibility, fire and structural protection of buildings, and environment).

In processing proposed changes or additions to any of the National Model Codes, the CCBFC considers many issues such as the following:

- Does the proposed requirement provide the minimum level of performance-and no more than the minimum-needed to achieve the Code's objectives?
- Will persons responsible for Code compliance be able to act on or implement the requirement using commonly accepted practices?
- Will enforcement agencies be able to enforce the requirement?

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- Are the costs of implementing the requirement justifiable?
 - Have the potential policy implications of the requirement been identified and addressed?
 - Is there broad consensus on this requirement among Code users representing all facets of the design and construction industries as well as among provincial and territorial governments?

Guidelines for requesting changes to the NBC are available on the NRC's website.

Objective-Based Code Format

The NBC has been published in an objective-based code format since 2005.

The 2006 BCBC (including plumbing services as Part 7) and the 2006 BCFC were the first objective-based codes adopted by the Province of British Columbia. The following relates also to Book II (Plumbing Systems), of this Code.

As described in more detail in the section entitled Structure of the NBC, the Code comprises three Divisions:

- Division A, which defines the scope of the Code and contains the objectives, the functional statements and the conditions necessary to achieve compliance;
- Division B, which contains acceptable solutions (commonly referred to as “technical requirements”) deemed to satisfy the objectives and functional statements listed in Division A; and
- Division C, which contains administrative provisions.

Most of the requirements in Division B are linked to three types of information:

- objectives (Code objectives, such as safety or health, which individual requirements help to address),
- functional statements (statements of the functions of the building that a particular requirement helps to achieve), and
- intent statements (detailed statements of the specific intent of the requirement).

Objectives

The NBC's objectives are fully defined in Section 2.2. of Division A.

The objectives describe, in broad terms, the overall goals that the NBC's requirements are intended to achieve. They serve to define the boundaries of the subject areas the Code addresses. However, the Code does not address all the issues that might be considered to fall within those boundaries.

The objectives describe undesirable situations and their consequences, which the Code aims to prevent from occurring in buildings. The wording of most of the definitions of the objectives includes two key phrases: “limit the probability” and “unacceptable risk.” The phrase “limit the probability” is used to acknowledge that the NBC cannot entirely prevent those undesirable situations from happening. The phrase “unacceptable risk” acknowledges that the NBC cannot eliminate all risk: the “acceptable risk” is the risk remaining once compliance with the Code has been achieved.

The objectives are entirely qualitative and are not intended to be used on their own in the design and approval processes.

The objectives attributed to the requirements or portions of requirements in Division B are listed in a table following the provisions in each Part of Division B.

Objectives have also been attributed to any requirements specific to British Columbia.

Functional Statements

The NBC's functional statements are listed in Section 3.2. of Division A.

The functional statements are more detailed than the objectives. They describe conditions in the building that help satisfy the objectives. The functional statements and the objectives are interconnected. There may be several functional statements related to any one objective, and a given functional statement may describe a function of the building that serves to achieve more than one objective.

Like objectives, functional statements are entirely qualitative and are not intended to be used on their own in the design and approval processes.

The functional statements attributed to the requirements or portions of requirements in Division B are listed in a table following the provisions in each Part of Division B.

Functional Statements have also been attributed to any requirements specific to British Columbia.

Intent Statements

Intent statements explain the basic thinking behind each Code provision contained in Division B. Intent statements, each of which is unique to the provision with which it is associated, explain how requirements help to achieve their attributed objectives and functional statements. Like the objectives, the intent statements are expressed in terms of risk avoidance and expected performance. They offer insight into the views of the responsible standing committees on what the Code provisions are intended to achieve.

The intent statements serve explanatory purposes only and do not form an integral part of the Code provisions. As such, they are similar in function to the explanatory Notes at the end of each Part. Due to the sheer volume of intent statements-thousands for the NBC alone-they are made available as a separate electronic document entitled "Supplement to the NBC 2020: Intent Statements," which is posted on the NRC's website.

"Supplement to the NBC 2020: Intent Statements" is not available at the time of printing of this Code. Intent statements have not been attributed to any requirements specific to British Columbia.

All this additional information-objectives, functional statements and intent statements-is intended to facilitate the implementation of the Code in two ways:

- **Clarity of intent:** The objectives, functional statements and intent statements linked to a Code requirement clarify the reasoning behind that requirement and facilitate understanding of what must be done to satisfy that requirement. This added information may also help avoid disputes between practitioners and officials over these types of issues.
- **Flexibility:** The additional information allows for flexibility in Code compliance. A person seeking to propose a new method or material not described or covered in the Code will be able to use the added information to understand the expected level of performance that their alternative solution must achieve to satisfy the Code.

Structure of the NBC

The NBC is organized into three Divisions, which are distributed across two volumes.

The remainder of this section relates also to Book II (Plumbing Systems) of this Code.

Division A: Compliance, Objectives and Functional Statements

Division A defines the scope of the NBC and presents the objectives that the Code addresses and the functions the building must perform to help to satisfy those objectives.

Division A cannot be used on its own as a basis for designing and constructing a building, or for evaluating a building's compliance with the Code.

Division B: Acceptable Solutions

The term “acceptable solutions” refers to the technical requirements contained in the Code. It reflects the principle that building codes establish an acceptable level of risk or performance and underlines the fact that a code cannot describe all possible valid design and construction options. Acceptable solutions represent the minimum level of performance that will satisfy the NBC's objectives and that is acceptable to an authority that adopts the NBC into law or regulation.

In context of the preceding paragraph, the phrase "authority that adopts the NBC into law or regulation" refers to the government of the Province of British Columbia. This differs from the context of application and enforcement of the BCBC, for which the Code contains a distinct definition of the term "authority having jurisdiction". (See Article 1.4.1.2. of Division A.)

Most of the requirements in Division B-the acceptable solutions-are linked to at least one objective and functional statement found in Division A. These linkages play an important role in allowing objective-based codes to accommodate innovation.

It is expected that the majority of Code users will primarily follow the acceptable solutions presented in Division B and that they will consult Division A only when seeking clarification on the application of Division B's requirements to a particular situation, when considering an alternative solution, or when looking up the definition of selected terms in the context of the NBC.

Division C: Administrative Provisions

Division C contains administrative provisions relating to the application of the Code. Many provinces and territories establish their own administrative provisions upon adopting or adapting the NBC; having all the administrative provisions in one Division facilitates their customization to suit jurisdictional needs.

The BCBC contains some unique administrative provisions in Division C.

In addition, a separate document entitled Administrative Requirements for Use with the National Building Code of Canada 1985 is automatically adopted, in accordance with Article 2.2.1.1. of Division C, if the authority having jurisdiction does not provide other administrative requirements.

Relationship between Division A and Division B

Sentence 1.2.1.1.(1) of Division A is a very important sentence: it is a precise statement of the relationship between Divisions A and B and is central to the concept of objective-based codes.

- 1)** Compliance with this Code shall be achieved by

 - a) complying with the applicable acceptable solutions in Division B (see Note A-1.2.1.1.(1)(a)), or

- b) using alternative solutions that will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions (see Note A-1.2.1.1.(1)(b)).

See Article 1.2.1.1. of the BCBC, which differs from above. Additional requirements apply in British Columbia.

Clause (a) makes it clear that the acceptable solutions in Division B are automatically deemed to satisfy the linked objectives and functional statements of Division A.

Clause (b) makes it clear that alternative solutions can be used in lieu of compliance with the acceptable solutions. However, to do something different from the acceptable solutions described in Division B, a proponent must show that their proposed alternative solution will perform at least as well as the acceptable solution(s) it is replacing. The objectives and functional statements attributed to the acceptable solution(s) identify the areas of performance where this equivalence must be demonstrated.

Parts in Division B and Professional Disciplines

Division B is organized into Parts that are largely related to professional disciplines. However, this does not mean that a person belonging to a certain profession who is executing the design or construction of a particular building component can necessarily work with only one Part of the Code in isolation since provisions related to that building component may be found in more than one Part.

For example:

- Provisions pertaining to fire safety issues related to heating, ventilating and air-conditioning (HVAC) systems are located in Part 3, Fire Protection, Occupant Safety and Accessibility, and not in Part 6, Heating, Ventilating and Air-conditioning.
- Structural requirements related to loads on handrails and grab bars are located in Part 3, Fire Protection, Occupant Safety and Accessibility, while structural requirements related to loads on guards and handrails are located in Part 4, Structural Design.

For this reason, the Part-based structure of Division B is not well suited for use as the basis for allocating responsibilities to different professions or as the basis for contractual arrangements.

Additional Information

Numbering System

A consistent numbering system has been used throughout the National Model Codes. The first number indicates the Part of the Code; the second, the Section in the Part; the third, the Subsection; and the fourth, the Article in the Subsection. The detailed provisions are found at the Sentence level (indicated by numbers in brackets), and Sentences may be broken down into Clauses and Subclauses. This structure is illustrated as follows:

3	Part
3.5.	Section
3.5.2.	Subsection
3.5.2.1.	Article
3.5.2.1.(2)	Sentence
3.5.2.1.(2)(a)	Clause
3.5.2.1.(2)(a)(i)	Subclause

Meaning of the Words “And” and “Or” between the Clauses and Subclauses of a Sentence

Multiple Clauses and Subclauses are connected by the word “and” or “or” at the end of the second last Clause or Subclause in the series. Although this connecting word appears only once, it is meant to apply to all the preceding Clauses or Subclauses within that series.

For example, in a series of five Clauses-(a) to (e)-in a Sentence, the appearance of the word “and” at the end of Clause (d) means that all Clauses in the Sentence are connected to each other with the word “and.” Similarly, in a series of five Clauses-(a) to (e)-in a Sentence, the appearance of the word “or” at the end of Clause (d) means that all Clauses in the Sentence are connected to each other with the word “or.”

In all cases, it is important to note that a Clause (and its Subclauses, if any) must always be read in conjunction with its introductory text appearing at the beginning of the Sentence. Moreover, the connecting words “and” and “or” must be read in the context of the Sentence. In particular, the use of the word “and” as a connecting word does not necessarily mean that all Clauses (or Subclauses) are applicable for compliance with the Sentence.

Change Indication

As a courtesy to Code users, efforts have been made to identify technical changes relative to the 2015 edition. Where a technical addition or revision has been made, a vertical line has been added in the margin next to the affected provision to indicate the approximate location of the new or revised content. No change indication is provided for editorial revisions or for renumbered or deleted content.

In addition to the above noted vertical lines, which indicate a change to the NBC, further unique to British Columbia technical changes, additions, and variations relative to the 2024 edition of the BCBC are identified. These changes are shown by a bold vertical line added in the margin next to the affected provision to indicate the approximate location of the unique to B.C. content, wherever practical. The bold vertical lines are for convenience only and have no legal effect. No change indication is provided for renumbered or deleted provisions. The term “reserved” is included in place of certain deleted Codes Canada model content which has not been adopted. The term “reserved” is generally used so that the numbering structure of the model National Codes is maintained, easing comparability and harmonization.

Units

All values in the NBC are given in metric units. Some of the metric values in the Code have been converted and rounded from imperial values. A conversion table of imperial equivalents for the most common units used in building design and construction is located at the end of the Code.

Complementary Publications

The following publications are referenced in the NBC 2020 or facilitate the application of its requirements:

- National Energy Code of Canada for Buildings 2020
- National Farm Building Code of Canada 1995
- National Fire Code of Canada 2020
- National Plumbing Code of Canada 2020
- Illustrated User's Guide – NBC 2020: Part 9 of Division B, Housing and Small Buildings
- Structural Commentaries (User's Guide – NBC 2020: Part 4 of Division B)
- Supplement to the NBC 2020: Intent Statements

These and other Code documents published by the NRC are made available in free electronic format on the NRC's website.

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Production and Marketing Manager
Codes Canada
National Research Council of Canada
1200 Montreal Road
Ottawa, Ontario K1A 0R6
E-mail: Codes@nrc-cnrc.gc.ca

Contact Information

The CCBFC welcomes comments and suggestions for improvements to the NBC. Persons interested in requesting a change to an NBC provision should refer to the guidelines available on the NRC's website.

To submit comments or suggestions, please contact:

The Secretary
Canadian Commission on Building and Fire Codes
1200 Montreal Road
Ottawa, Ontario K1A 0R6
E-mail: CCBFCSecretary-SecretaireCCCBPI@nrc-cnrc.gc.ca

Relationship of the NBC to Standards Development and Conformity Assessment

The development of many provisions in the NBC and the assessment of conformity to those provisions are supported by several of the member organizations of Canada's National Standards System (NSS).

The NSS is a network of accredited organizations concerned with standards development, certification, testing and inspection that is established under the auspices of the Standards Council of Canada Act. Activities of the NSS are coordinated by the Standards Council of Canada (SCC), which accredits standards development organizations, certification bodies, testing and calibration laboratories, and inspection bodies, among others.

The SCC is a non-profit federal Crown corporation responsible for the coordination of voluntary standardization in Canada. It also coordinates Canadian participation in voluntary international standardization activities.

The remainder of this section relates also to Book II (Plumbing Systems) of this Code

Canadian Standards

Many of the standards referenced in the NBC are published by standards development organizations accredited in Canada. As part of the accreditation requirements, these organizations adhere to the principle of consensus, which generally means substantial majority agreement of a committee comprising a balance of producer, user and general interest members, and the consideration of all negative comments. The standards development organizations also have formal procedures for the balloting and second-level review of standards prepared under their oversight.

The following organizations are accredited as standards development organizations in Canada:

- Air-Conditioning, Heating and Refrigeration Institute (AHRI)
- ASTM International
- Bureau de normalisation du Québec (BNQ)
- Canadian General Standards Board (CGSB)
- CSA Group
- International Association of Plumbing and Mechanical Officials (IAPMO)
- ULC Standards
- Underwriters' Laboratories Inc. (UL)

Tables 1.3.1.2. and D-1.1.2. of Division B list the standards referenced in the NBC. Standards proposed to be referenced in the NBC are reviewed to ensure that their content is compatible with the Code. Thereafter, referenced standards are reviewed as needed during each Code cycle. Standards development organizations are asked to provide information on any changes in the status of their standards referenced in the NBC—withdrawals, amendments, new editions, etc. This information is passed on to the CCBFC, its standing committees, the provinces and territories, and interested stakeholders, all of whom are given the opportunity to identify any problems associated with the changes. These bodies do not necessarily review in detail the revised standards; rather, the approach relies on the consensus process involved in the maintenance of the standards and on the extensive knowledge and experience of committee members, provincial or territorial staff, NRC staff, and consulted stakeholders to identify changes in the standards that might create problems in the Code.

Non-Canadian Standards

A number of subject areas for which the standards development organizations accredited in Canada have not developed standards are covered in the NBC. In these cases, the Code often references standards developed by organizations in other countries, such as the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) and the National Fire Protection Association (NFPA). These standards are developed using processes that may differ from those used by the standards development organizations accredited in Canada; nevertheless, the standards have been reviewed by the relevant standing committees and found to be acceptable.

Conformity Assessment

The NBC establishes minimum measures, which are set out within its own text or within referenced standards. However, the NBC does not set out who is responsible for assessing conformity to the measures or how those with this responsibility might carry it out. This responsibility is usually established by the governing legislation of the adopting provinces and territories. Provincial or territorial authorities should be consulted to determine who is responsible for conformity assessment within their jurisdiction.

In British Columbia, local authorities are authorized to, by bylaw, regulate construction and to enforce the requirements of the BCBC.

Those persons responsible for ensuring that materials, appliances, systems and equipment meet the requirements of this Code have several means available to assist them, ranging from on-site inspection to the use of certification services provided by accredited third-party organizations. Test reports or mill certificates provided by manufacturers or suppliers can also assist in the acceptance of products. Engineering reports may be required for more complex products.

Requirements for Registered Professionals are located in Division C of the BCBC

Testing

The SCC accredits testing and calibration laboratories that are capable of reliably testing products to specified standards. The test results produced by these organizations can be used in the certification, evaluation and qualification of products for compliance with Code provisions. The SCC's website (www.scc.ca) lists accredited testing and calibration laboratories, along with their scope of accreditation.

Certification

Certification is the confirmation by an independent organization that a product, process, service or system meets a requirement. Certification may entail physical examination, testing as specified in appropriate standards, an initial plant inspection, and/or follow-up unannounced plant inspections. This procedure leads to the issuing of a formal assurance or declaration, by means of a certification mark or certificate, that the product, process, service or system is in full conformity with specified provisions.

In some cases, a product for which no standard exists can be certified using procedures and criteria developed by an accredited certification body and specifically designed to measure the performance of that product.

Certification bodies publish lists of certified products and companies. The SCC's website (www.scc.ca) lists accredited certification bodies, along with their scope of accreditation. Several organizations, including the Canadian Construction Materials Centre (CCMC) at the NRC, offer product certification services.

Evaluation

An evaluation is a written opinion by an independent professional organization that a product will perform its intended function. An evaluation is often done to determine the ability of an innovative product, for which no standards exist, to satisfy the intent of a Code requirement. Follow-up plant inspections are not normally part of the evaluation process.

Qualification

Qualification evaluates the ability of a product to perform its intended function by verifying that it meets the requirements of a standard. Qualification normally includes some follow-up plant inspection. Some organizations publish lists of qualified products that meet the specified requirements. Some organizations qualify product manufacturing and/or testing facilities for compliance with the Code and relevant standards.